LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY
SIERRA LEONE

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOG

PROG101 PROGRAMMING LOGIC

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ABSTRACT

This assignment examines unreliable electricity access in rural Sierra Leone and proposes an open-source Community Energy Service Platform to help residents report power outages and track repairs to local mini-grids. The system's logical design uses computational thinking, modular processes, algorithms, pseudocode, and flowcharts to show how reports can be received, prioritized, assigned, and resolved. Designed for limited internet access, it aims to improve communication between communities and energy providers. Open-source development and Digital Public Goods principles can support local adaptation, accessibility, and shared improvement. The proposal contributes to **Sustainable Development Goal 7: Affordable and Clean Energy** by supporting more reliable and inclusive electricity services.

INTRODUCTION

Reliable electricity is essential for schools, clinics, businesses, and households, yet access in Sierra Leone remains limited, particularly in rural communities. Power interruptions and unclear ways to report faults can disrupt learning, health services, and local livelihoods.

This proposal recommends the Sierra Leone Community Energy Service Platform (SCESP), an accessible, offline-first tool for reporting outages, safety concerns, and billing problems, while helping operators track repairs and monitor mini-grid performance. Its open-source design would allow Sierra Leonean communities, developers, and institutions to contribute improvements and adapt it to local needs.

SCESP would complement investments in electricity infrastructure by strengthening communication, maintenance, and accountability. Its long-term aim is to support more reliable and inclusive energy services in Sierra Leone, contributing to **SDG 7: Affordable and Clean Energy**.

Keeping Community Power Reliable

An open-source service platform for Sierra Leone's rural mini-grids

A practical proposal aligned with Sustainable Development Goal 7

EXECUTIVE SUMMARY

Sierra Leone should complement rural renewable-energy investment with an offline-first, open-source platform that helps communities report outages, helps operators track repairs, and makes service performance visible. The platform will strengthen maintenance and accountability; it cannot generate electricity or replace investment in physical infrastructure.

1. The problem

Electricity access remains limited and uneven, while a connection alone does not guarantee dependable service. The World Bank's Sierra Leone energy access diagnostic found that 21.1% of households had electricity through the national grid or mini-grids and a further 14.7% through off-grid sources. These are household survey figures and should not be directly compared with government coverage estimates that may use different definitions. The consistent concern is that rural and remote communities remain underserved and service reliability and affordability matter alongside connection rates.

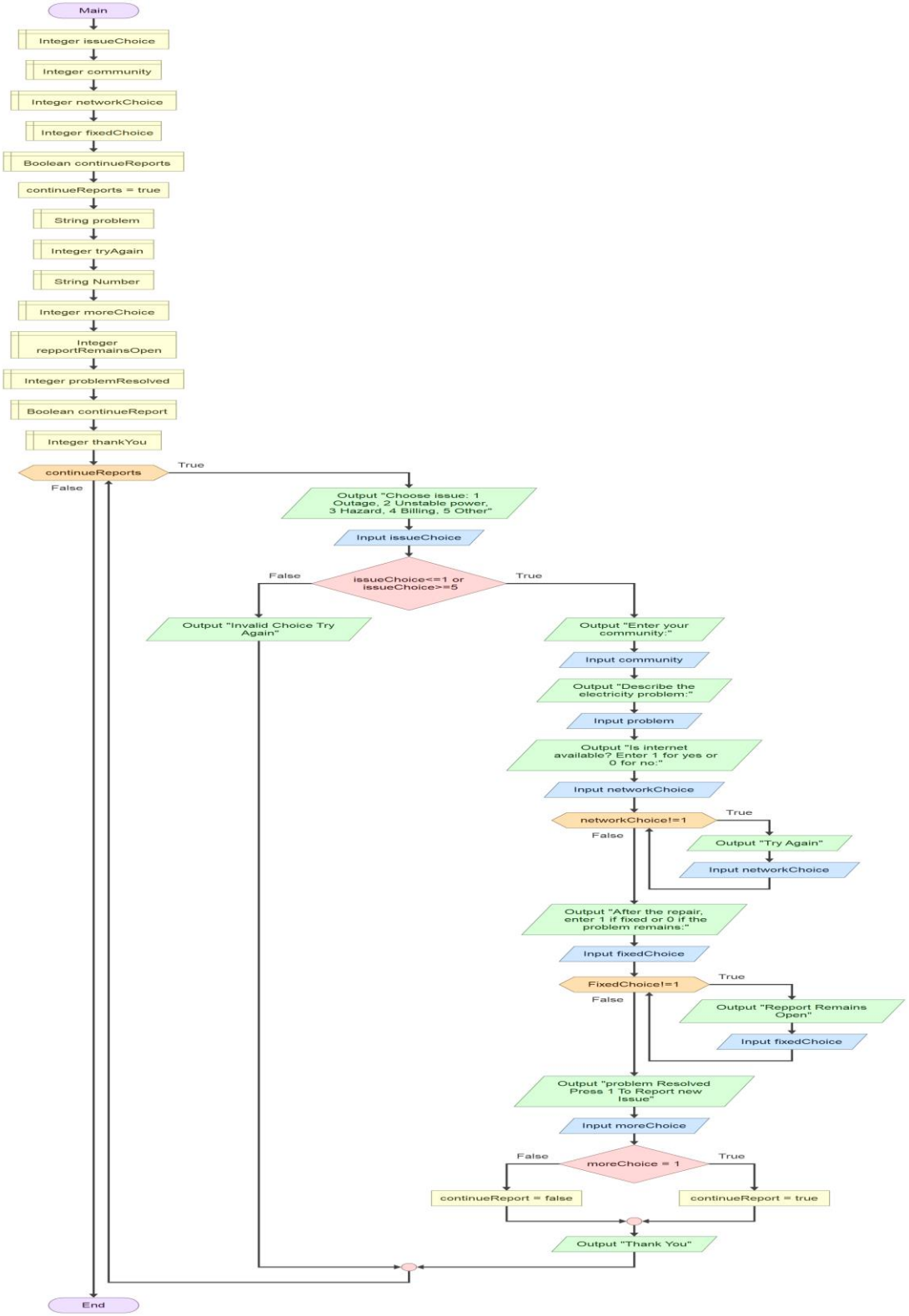
The effects reach beyond lighting. Clinics can struggle to maintain essential services, students lose study time, and small businesses, farms, and fishing communities lose productive hours or goods when electricity is unavailable. Sierra Leone has an established mini-grid base: a World Bank review reported 104 mini-grids, about 15,000 connections, and roughly 5 MW of capacity in its data snapshot. As systems expand, communities and operators need a straightforward way to identify faults and follow repair requests through

2. Proposed solution

Create the Sierra Leone Community Energy Service Platform (SCESP), a shared public-interest digital service for participating mini-grid communities. It would connect residents, local technicians, operators, district authorities, and national energy institutions through a simple report-to-repair process.

- **Report a problem:** Residents submit a blackout, safety issue, billing concern, or service complaint using a lightweight mobile web form, SMS/USSD pathway where feasible, or a trained local focal person. Each report receives a reference number.
- **Track the response:** Operators receive a work queue, acknowledge issues, record repairs, and add notes on recurring faults and parts. Communities can check whether a report is open, in progress, or resolved.
- **Show service trends:** A public dashboard presents aggregate uptime, outage duration, response time, tariffs, and unresolved reports by service area. It excludes names, phone numbers, and other personal information.
- **Work with weak connectivity:** The field tool stores reports offline and synchronizes when a connection returns. SMS and assisted reporting provide options for people without smartphones or data.

FLOWCHART


Main
Integer issueChoice
Integer community
Integer networkChoice
Integer fixedChoice
Boolean continueReports
continueReports = true
String problem
Integer tryAgain
String Number
Integer moreChoice
Integer repportRemainsOpen
Integer problemResolved
Boolean continueReport
Integer thankYou
continueReports
True
False
Output "Choose issue: 1 Outage, 2 Unstable power, 3 Hazard, 4 Billing, 5 Other"
Input issueChoice
issueChoice<=1 or issueChoice>=5
False
True
Output "Invalid Choice Try Again"
Output "Enter your community:"
Input community
Output "Describe the electricity problem:"
Input problem
Output "Is internet available? Enter 1 for yes or 0 for no:"
Input networkChoice
networkChoice!=1
True
False
Output "Try Again"
Input networkChoice
Output "After the repair, enter 1 if fixed or 0 if the problem remains:"
Input fixedChoice
FixedChoice!=1
True
False
Output "Repport Remains Open"
Input fixedChoice
Output "problem Resolved Press 1 To Report new Issue"
Input moreChoice
moreChoice = 1
False
True
continueReport = false
continueReport = true
Output "Thank You"
End

3. Expected impact and beneficiaries

The near-term result should be better visibility of service failures and clearer responsibility for resolving them. Traceable reports can help reduce the time problems remain unseen, reveal repeat faults, and help public agencies and funders target maintenance. The platform cannot by itself guarantee fewer outages; its value depends on operators having the capacity and resources to act on reports.

WHO BENEFITS	HOW THEY BENEFIT
Households and residents	A clear, traceable route to report service problems, including rural residents and people often missed by digital services.
Clinics and schools	More visible outages and a documented channel to raise issues affecting essential services.
Small businesses and livelihoods	Operators can see recurring disruption affecting traders, farms, fishing, and local enterprise.
Operators and public agencies	Structured repair histories and comparable service data to prioritize maintenance and future investment.

Measure success against a baseline: outage frequency and duration, acknowledgement and repair times, repeat faults, share of reports resolved, and user-reported ease of reporting. Publish coverage and methods so the numbers are interpretable. Collect only the personal data needed to respond, and disaggregate feedback only when it is safe and useful.

4. Why open source and how the community can contribute

Because SCESP would support a shared public service, its core code should use an OSI-approved license such as Apache-2.0. Open code can be inspected, adapted, and reused by Sierra Leonean institutions and local developers. It can make vendor choices more contestable, reduce dependence on one supplier, and let other energy programs reuse components. These are opportunities, not automatic savings or security guarantees: the service still needs funded maintenance, local technical skills, security review, and clear ownership.

- **Community contributions:** Translate and test plain-language messages; check the service with rural residents and people with disabilities; document local reporting needs; report bugs and suggest features; validate public facility and service-area records; and train local technicians.

Shared stewardship: Publish contribution guidelines, a public issue tracker, release notes, and decisions from a local steering group that includes community and operator representatives.

5. Accessibility, openness, and public benefit

Accessibility: Design for intermittent connectivity and basic phones, provide assisted reporting and local-language options, and test screen-reader use, keyboard navigation, contrast, and text resizing against WCAG 2.2 with people with disabilities.

Openness: Publish source code, setup and maintenance instructions, data definitions, governance decisions, and interoperable APIs. Use open formats so authorized users can move or reuse data without being locked to one vendor.

Public benefit: Give residents a free way to report problems and see aggregate service performance. Protect complainants and avoid publishing sensitive infrastructure details; provide a clear escalation path for unresolved cases.

6. Path to becoming a digital public good

Open source alone does not make SCESP a digital public good (DPG). The

Digital Public Goods Alliance Standard also considers SDG relevance, open licensing, clear ownership, platform independence, documentation, non-personal data extraction, privacy and legal safeguards, and security and harm prevention. SCESP should publish its governance and data policies, document its APIs, use open standards, and keep public datasets free of personal information. Once it is mature and meets the requirements, maintainers can submit it for independent review and possible inclusion in the DPG Registry; recognition is not automatic.

7. SDG alignment and long-term goal

SDG 7 | Affordable and Clean Energy

The proposal supports SDG 7, especially Target 7.1: universal access to affordable, reliable, and modern energy services by 2030. Better outage reporting and repair tracking support reliability and accountability, while the renewable mini-grids provide the physical energy service.

Start with a co-designed pilot in a small number of mini-grid communities. Agree who receives reports, response targets, escalation steps, and baseline indicators before launch. Test offline use and accessibility, publish progress, and expand only when communities find the service useful and operators can respond. Align the work with Sierra Leone's Mission 300 energy compact and existing rural electrification programs to avoid duplicate systems.

The long-term goal is for participating communities to have a dependable feedback and maintenance loop, supported by Sierra Leonean institutions and contributors. This would strengthen the reliability and public accountability of energy access while contributing to affordable, reliable, sustainable, and modern energy for all.

ALGORITHM

Open SCESP.

2. Choose the issue.

3. Enter the location/community

4. Submit the report

5. Get a reference number.

6. Wait for the operator’s response.

7. Check progress.

8. Confirm the outcome.

PSEUDOCODE

START

1. OPEN SCESP
2. DISPLAY "Choose the issue"
3. INPUT issue
4. DISPLAY "Enter location/community"
5. INPUT location
6. SUBMIT issue, location
7. GENERATE reference_number
8. DISPLAY reference_number
9. WAIT FOR operator_response
10. CHECK progress
11. IF issue is resolved THEN
    * DISPLAY "Confirm the outcome"
    * INPUT confirmation
12. ELSE
    * DISPLAY "Continue waiting for a response"
13. ENDIF

STOP

CONCLUSION

Sierra Leone's electricity challenge requires more than building new mini-grids. Communities also need a reliable way to report failures, track repairs, and hold service providers accountable. The proposed Sierra Leone Community Energy Service Platform would create that connection through an accessible, offline-first, open-source system.

By involving residents, local technicians, operators, regulators, and public institutions, the platform could improve repair response, strengthen trust, support evidence-based investment, and make electricity services more inclusive. Its open-source design would allow Sierra Leonean developers and communities to adapt, maintain, and expand the system without dependence on one private supplier.

In the long term, SCESP could become a digital public good that supports Sierra Leone's energy goals and contributes to SDG 7: ensuring access to affordable, reliable, sustainable, and modern energy for all.

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